MAXIMIZING REVENUE FOR DRIVERS

Through Payment Type

Optimizing Fare Pricing & Trip Distances via Payment Method Analysis

DATASET CONTEXT

NYC Taxi and Limousine Commission (TLC) Trip Records

Agenda

- Problem Statement
- Research Question
- Data Overview
- Methodology
- Analysis and Findings
- Hypothesis Testing
- Recommendations

Problem Statement

In the fast-paced taxi booking sector, making the most of revenue is essential for **long-term success and driver happiness**.

Our goal is to use data-driven insights to **maximise revenue streams** for taxi drivers in order to meet this need.

Our research aims to determine whether payment methods have an impact on fare pricing by focusing on the relationship between payment type and fare amount.

Research Question

Is there a relationship between total fare amount and payment type?

Can we nudge customers towards payment methods that generate higher revenue for drivers, without negatively impacting customer experience?

+18.9%
OBSERVED CARD FARE PREMIUM

+22.0%
LONGER TRANSIT DISTANCE

82.9%
CURRENT DIGITAL ADOPTION

Data Overview

For this analysis, we utilized the comprehensive dataset of NYC Taxi Trip records, used data cleaning and feature engineering procedures to concentrate solely on the relevant columns essential for our investigation.

Relevant columns used for this research:

- passenger_count (1 to 6)
- payment_type (Card or Cash)
- fare_amount (base fare $)
- trip_distance (miles)
- duration (minutes)

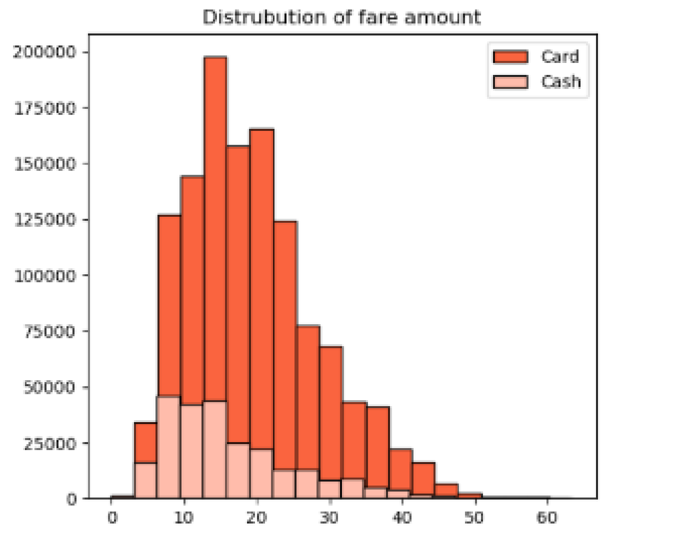
passenger_count	trip_distance	payment_type	fare_amount	duration
1	7.51 mi	Cash	$35.90	27.82 min
1	6.14 mi	Card	$27.50	22.70 min
1	2.40 mi	Card	$19.10	19.95 min
2	0.86 mi	Card	$7.20	5.30 min
1	5.89 mi	Card	$28.90	25.33 min

Methodology

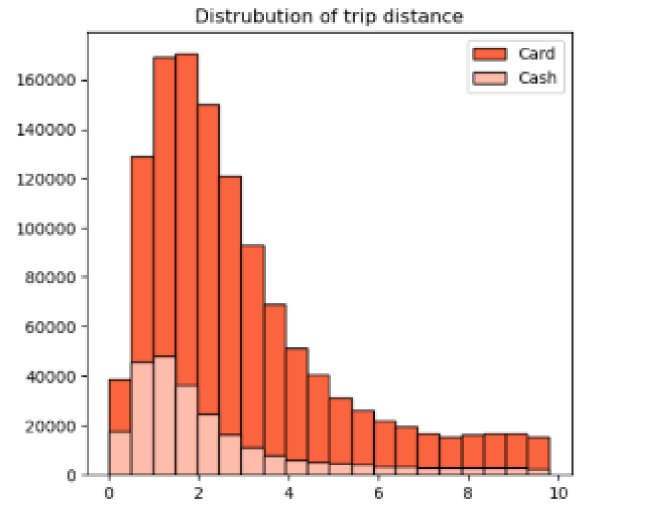
Step	Description
Data Cleaning & Hygiene	Derived trip duration; resolved 955K missing values (proved 100% Flex Fares); pruned deduplicated rows; removed non-positive values and capped outliers via 1.5×IQR boundary trimming.
Descriptive Analysis	Performed statistical analysis to summarize key aspects of the data, focusing on fare amounts, transit distance, and payment types across groups.
Confounder Analysis	Cross-tabulated passenger volume against payment channels to prove group capacity does not bias the fare differences between Card and Cash.
Hypothesis Testing	Conducted Welch's Two-Sample Independent T-test to evaluate the relationship between payment type and fare amount, statistically testing if differences are significant.

Journey Insights

- Customers paying with cards tend to have a slightly higher average trip distance and fare amount compared to those paying with cash.
- Indicates that customers prefer to pay more with cards when they have high fare amount and long trip distance.



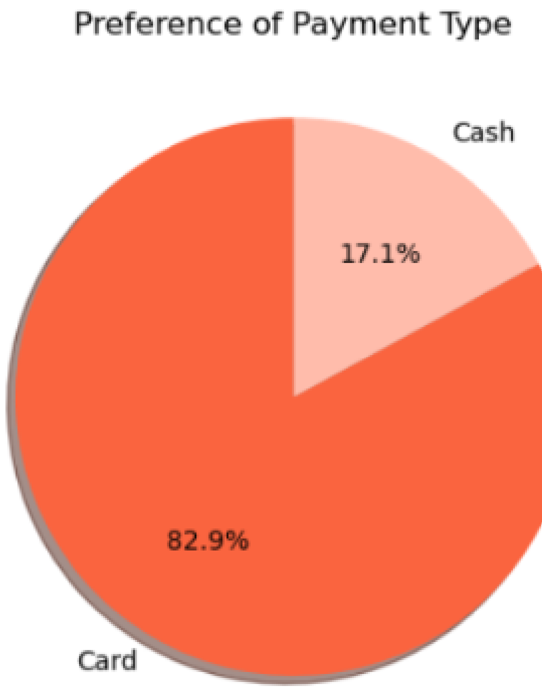
Fare Amount Distribution



Trip Distance Distribution

Metric	Payment Type	Mean	Standard Deviation
Fare amount	Card	$19.44	$9.20
Cash	$16.35	$9.11	
Trip distance	Card	2.94 mi	2.16 mi
Cash	2.41 mi	2.14 mi	

Preference of Payment Types



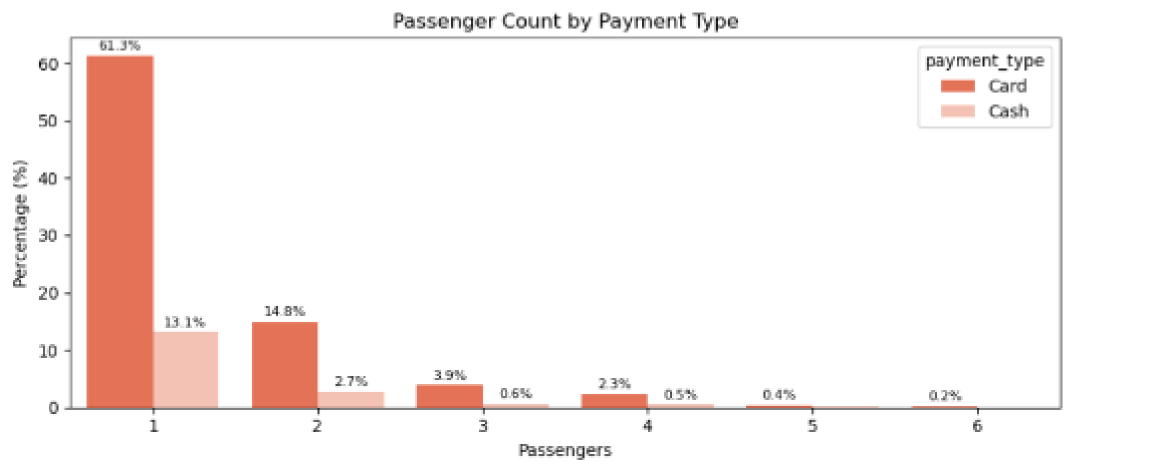
The proportion of customers paying with cards is significantly higher than those paying with cash, with **card payments accounting for 82.9%** of all transactions compared to cash payments at **17.1%**.

This indicates a strong preference among customers for using card payments over cash, potentially due to convenience, security, or corporate expense habits.

+$3.09 per ride

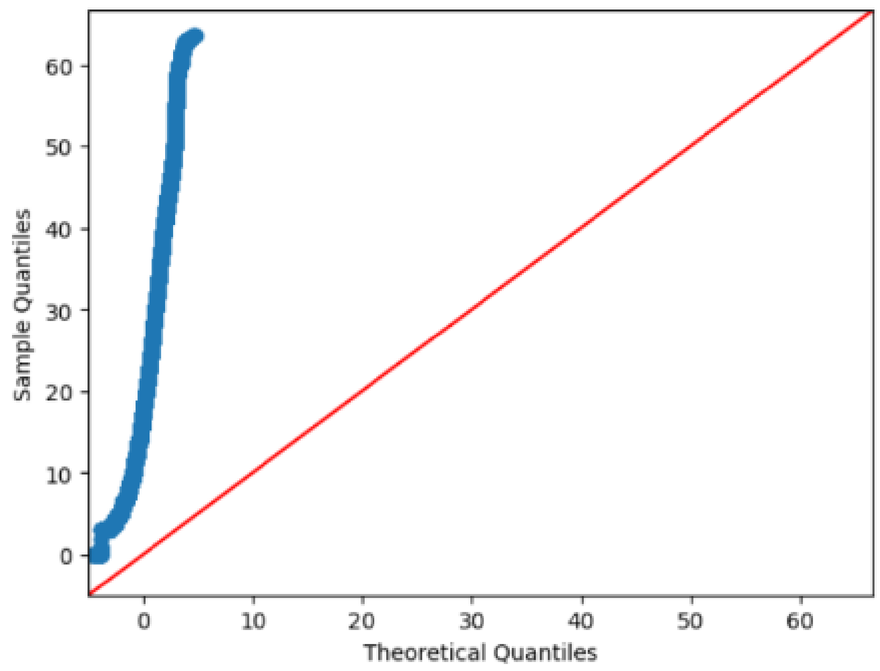
DRIVER REVENUE LIFT ON CARD TRANSACTIONS (+18.9%)

Passenger Count Analysis



- Among card payments, rides with a single passenger (passenger_count = 1) comprise the largest proportion, constituting **61.3% of all card transactions** (and 13.1% for cash).
- Similarly, two-passenger rides represent 14.8% Card vs. 2.7% Cash, while larger groups (3 to 6) decrease rapidly.
- There is a noticeable decrease in the percentage of transactions as passenger count increases, with 5 and 6 passenger trips accounting for under **1%** of total volume.
- **Key Confounder Takeaway:** Payment preference ratios remain fixed (~5:1) across all party sizes, confirming party size does not bias fare differences.

Hypothesis Testing



Q-Q Plot: CLT ensures mean normality ($N > 1.48M$)

Null hypothesis (H_0): There is no difference in average fare between customers who use credit cards and customers who use cash.

Alternative hypothesis (H_a): There is a difference in average fare between customers who use credit cards and customers who use cash.

Welch's Two-Sample T-Test Output

T-STATISTIC

t = 155.24

OBSERVED P-VALUE

p = 0.0000

With a T-statistic of 155.24 and a P-value far below 0.05, we **decisively reject the null hypothesis**. There is indeed a statistically significant difference in average fare between payment channels.

Recommendations

Encourage customers to pay with credit cards to capitalize on the potential for generating more revenue for taxi cab drivers.

Implement strategies such as offering incentives or discounts for credit card transactions to incentivize customers to choose this payment method.

Provide seamless and secure credit card payment options to enhance customer convenience and encourage adoption of this preferred payment method.

Image Sources

https://storage.ghost.io/c/3e/3b/3e3b259f-e5a2-4287-97e8-4888d22daaef/content/images/wp-content/uploads/2022/10/marvelous-mrs-maisel-filming-locations-horn-and-hardart-set-amazon-studios-crown-heights-brooklyn-nyc14.jpg

Source: www.untappedcities.com

https://images.stockcake.com/public/a/d/6/ad6548c1-6114-462c-8769-404c126151fa_large/golden-success-arrow-stockcake.jpg

Source: stockcake.com